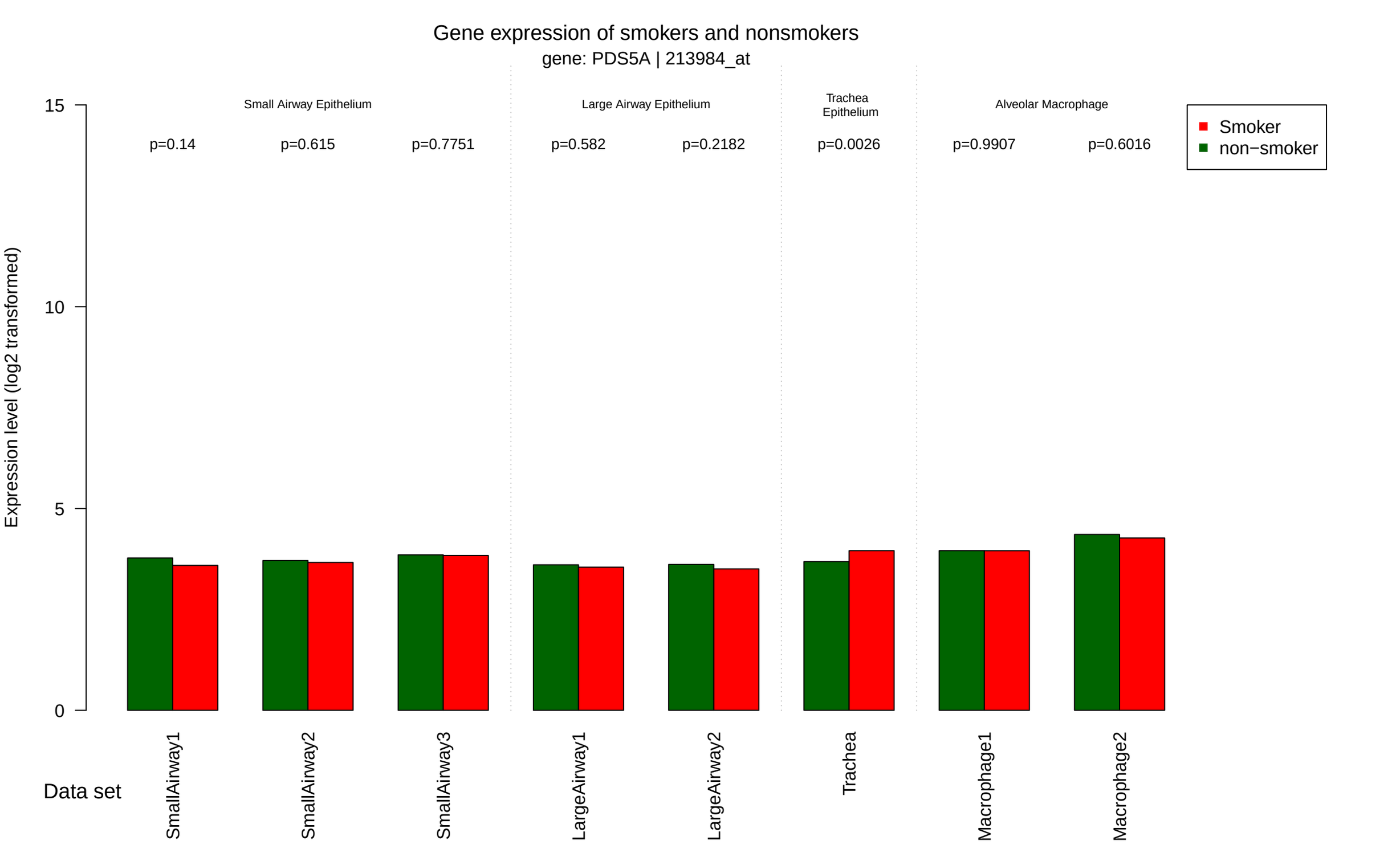
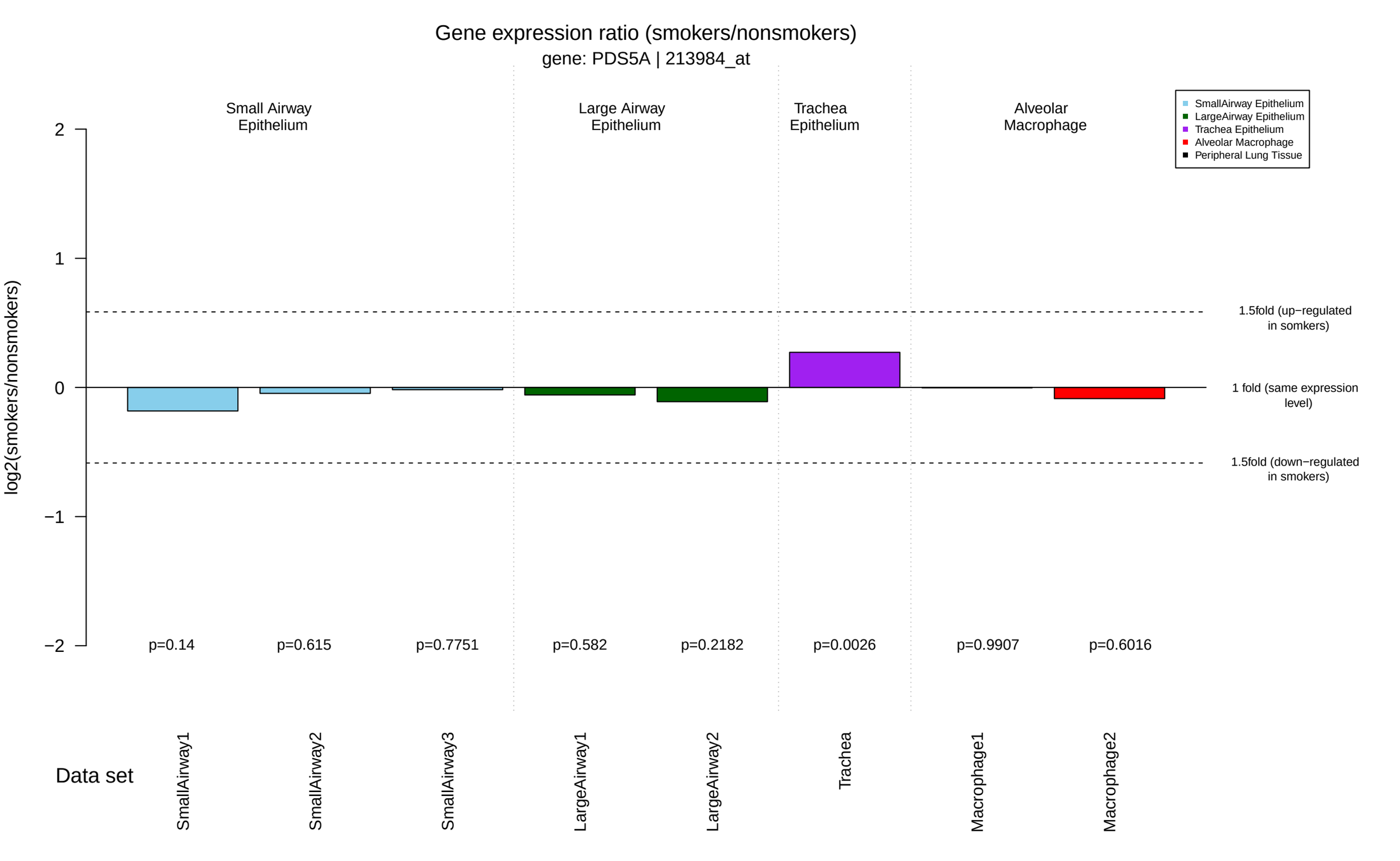
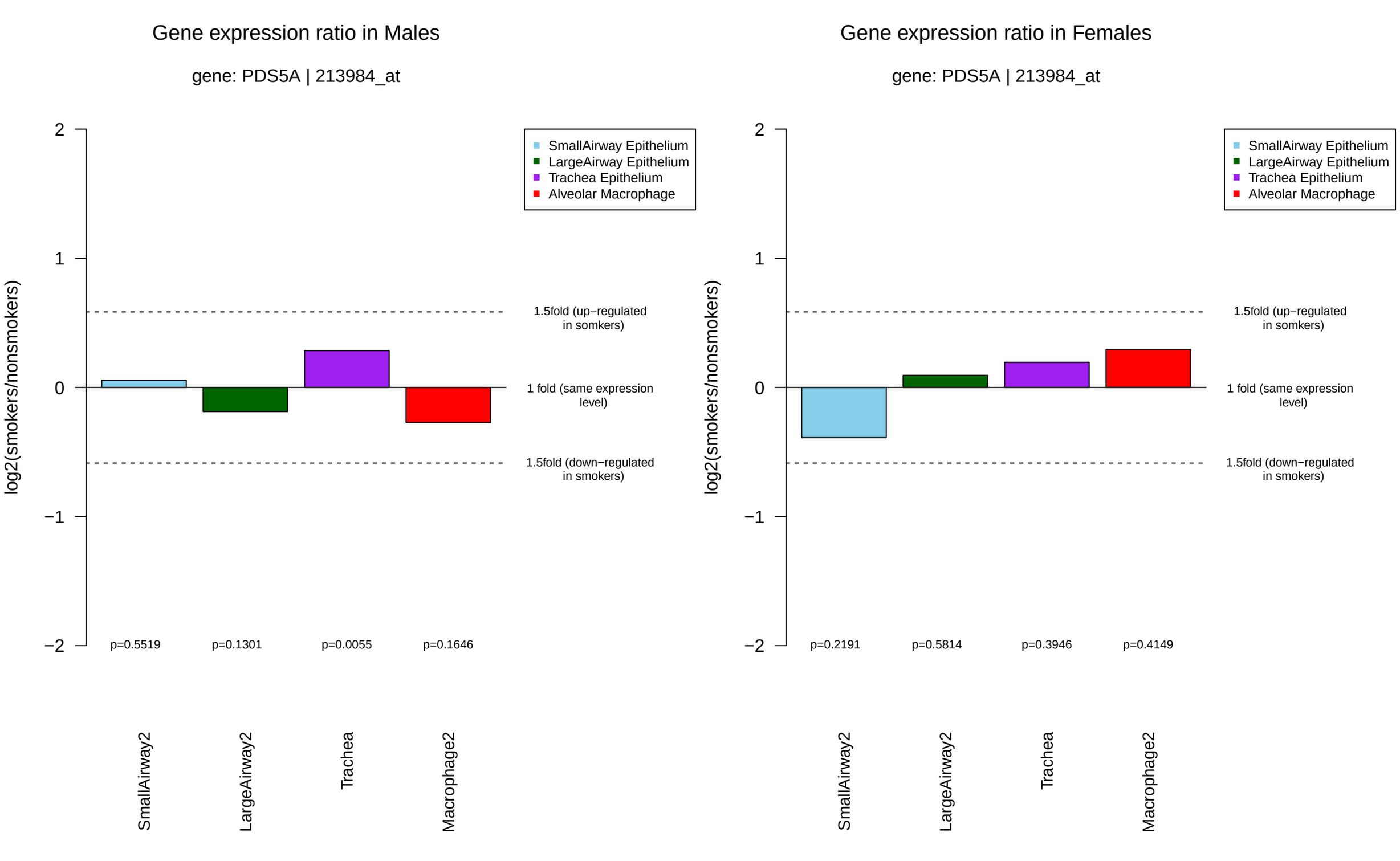
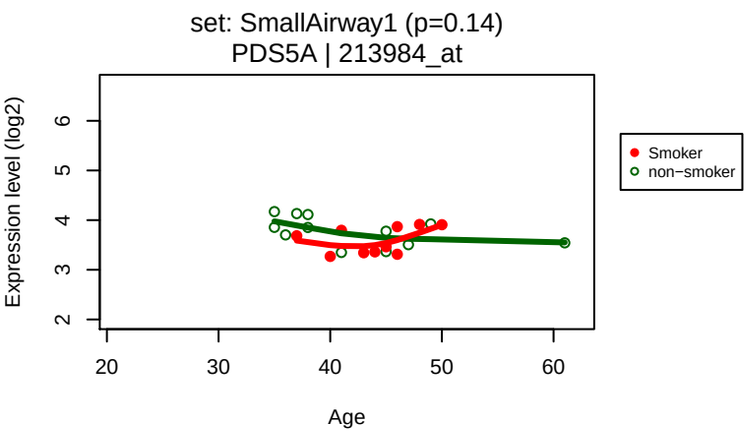
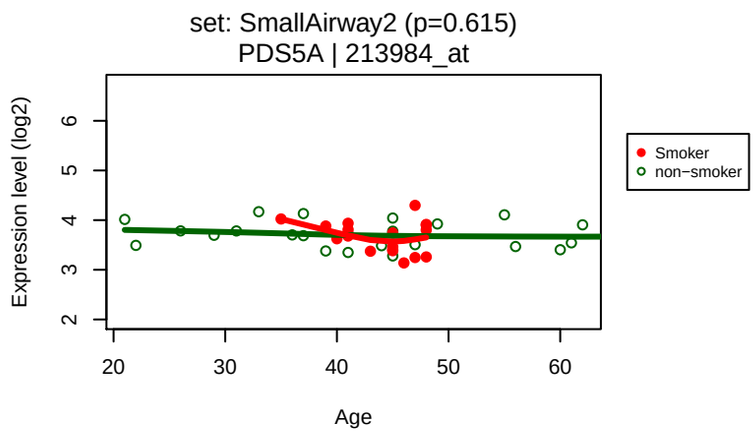
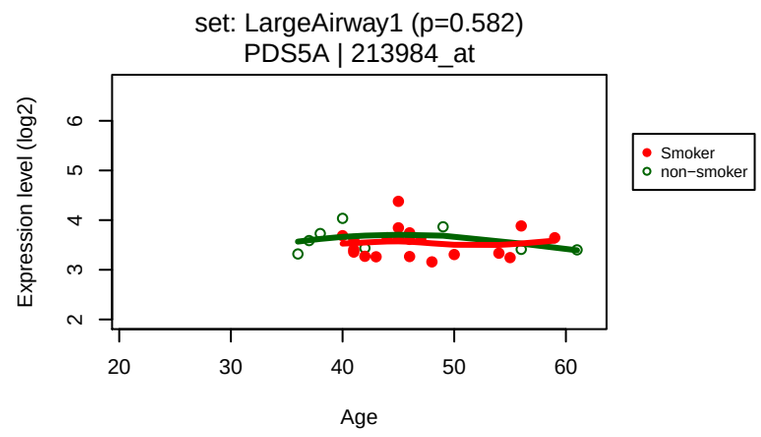

Gene expression of smokers and nonsmokers
gene: PDS5A | 213984_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.14
p=0.615
p=0.7751
p=0.582
p=0.2182
p=0.0026
p=0.9907
p=0.6016
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

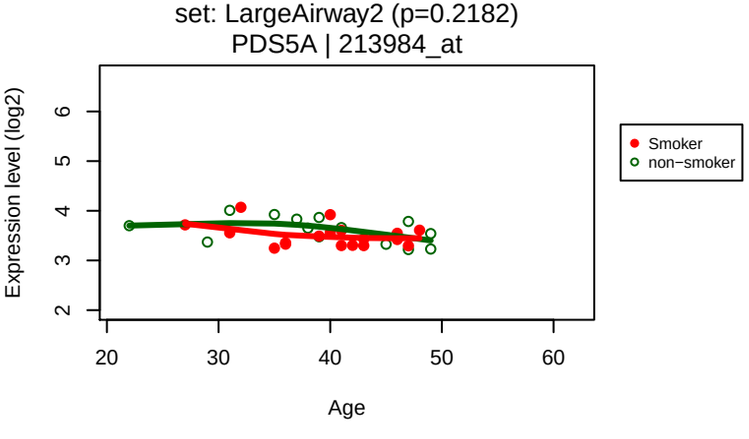

Gene expression ratio (smokers/nonsmokers)
gene: PDS5A | 213984_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.14
p=0.615
p=0.7751
p=0.582
p=0.2182
p=0.0026
p=0.9907
p=0.6016
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

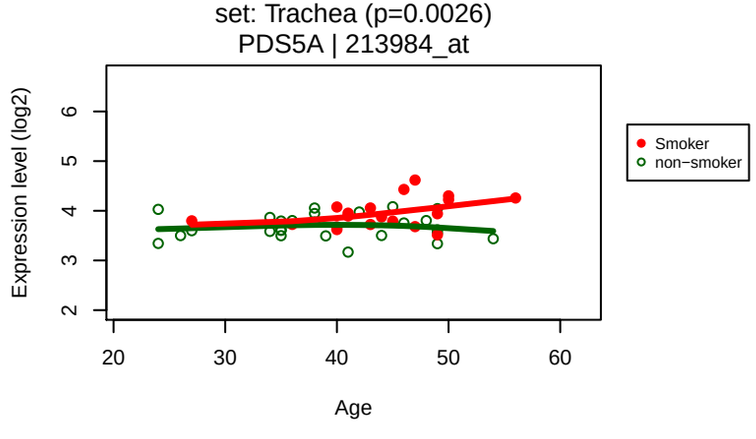

Gene expression ratio in Males
gene: PDS5A | 213984_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5519
p=0.1301
p=0.0055
p=0.1646
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDS5A | 213984_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2191
p=0.5814
p=0.3946
p=0.4149
SmallAirway2
LargeAirway2
Trachea
Macrophage2

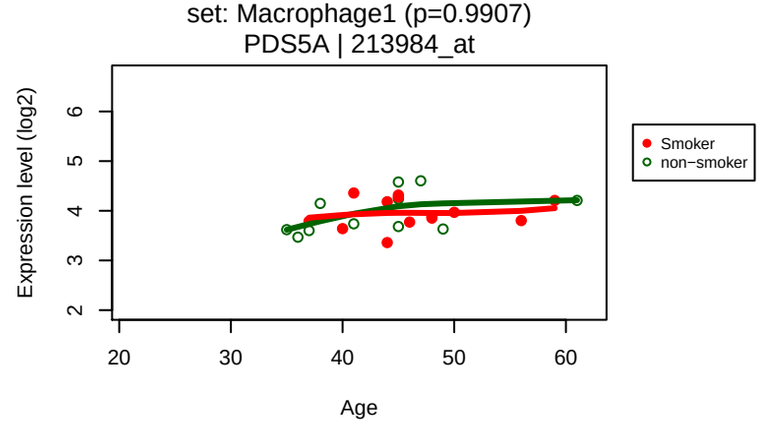

set: SmallAirway1 (p=0.14)
PDS5A | 213984_at
Expression level (log2)
Age
Smoker
non−smoker

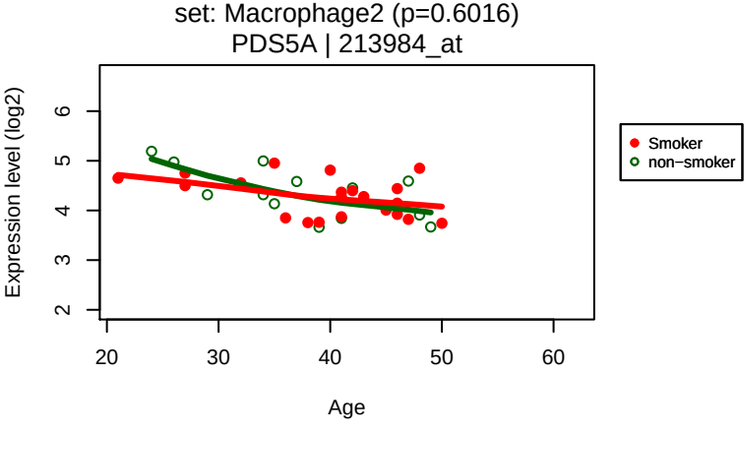

set: SmallAirway2 (p=0.615)
PDS5A | 213984_at
Expression level (log2)
Age
Smoker
non−smoker

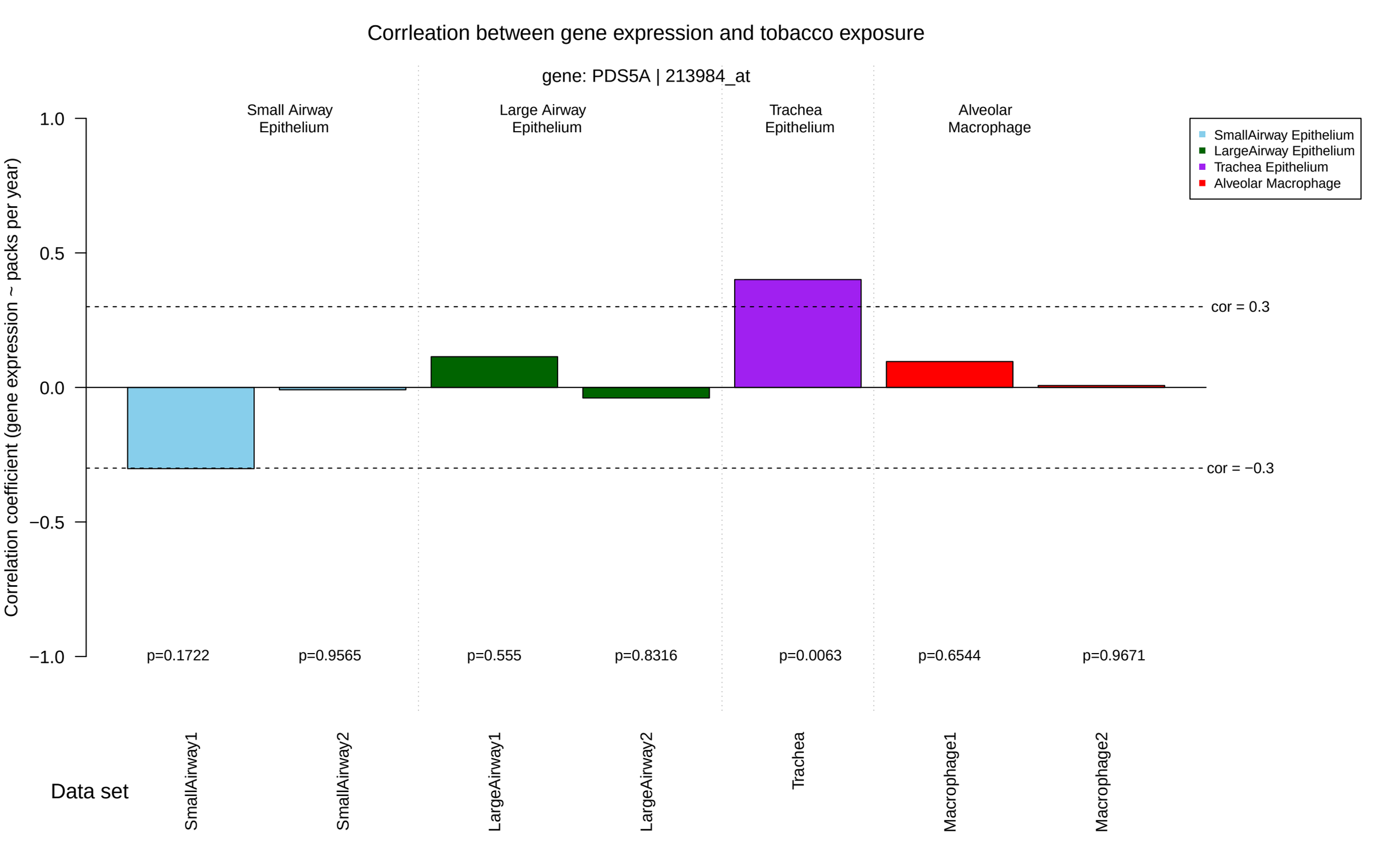

set: LargeAirway1 (p=0.582)
PDS5A | 213984_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2182)
PDS5A | 213984_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0026)
PDS5A | 213984_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9907)
PDS5A | 213984_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6016)
PDS5A | 213984_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDS5A | 213984_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1722
p=0.9565
p=0.555
p=0.8316
p=0.0063
p=0.6544
p=0.9671
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2